Depth bought the MLP back

N=48, degree 5, 4 colours, leak-free | stage E at E1 3/3, E2 3/3 seeds | 78 cells

THE SECOND AXIS, WITH A PAIRED CONTROL AT IDENTICAL DEPTH

Stage D handed an attention-only model more WIDTH than the hub and it did not cross. Stage E hands it more DEPTH: no MLP anywhere, at L=8 and at L=12, against a hub at L=6. Both depths were picked because the COMPLETE model is already measured there on this exact grid and budget, so neither rung is compared to an extrapolation -- full L=8 sits at 0.471 and full L=12 at 0.385, and attention-only at the hub's own depth sits at 0.763.

WHAT THE PRE-REGISTERED SCORER RETURNS

E1, no_mlp L=8: 0.749 / 0.3214 at 2,274,874 trainable, against the complete model at the SAME depth, 0.471 / 0.7111 at 6,479,418; plateau escapes 3/3, 3 of 3 seeds crossing -- HELD. E2, no_mlp L=12: 0.605 / 0.5214 at 3,331,642 trainable, against the complete model at the SAME depth, 0.385 / 0.8163 at 9,638,458; plateau escapes 3/3, 3 of 3 seeds crossing -- FALSIFIED. The two calls are printed in full on page 6, in the scorer's own words.

AND THE WAY IT PRODUCES NOTHING WAS NAMED FIRST

Attention-only escapes finding 44's 1.64 plateau on 2 of 3 seeds at L=6 and 0 of 3 at L=4, so a DEEPER attention-only stack failing to escape is the EXPECTED failure mode, written into the header before the job ran and scored here as 'did not train in 375k steps' -- never as evidence that the barrier holds. The budget was deliberately not raised to rescue it: a rung trained on a different budget cannot be read against the ladder it is compared to. Stage G (51194118) moves that knob on purpose, as its own arm.

rho_50 = realized hint density at which solve rate crosses 0.5 on a 14-point x 400-instance grid, LOWER IS BETTER, reported CENSORED when no bin crosses. mean_solve = unweighted mean over the 14 bins, which never censors. The seed-spread bar s* = 0.1484 / 0.2918 was frozen before any of these cells ran. Commit c04ecc9.

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING
	BEGIN_ROUND SOLVED UNRESOLVED
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3
	the q colours (token name COLOR_n, prose 'colour')
10-57	N0 N1 ... N47
	one id per node, no structure in the id

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*

SOLVED

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 401 TOKENS

HINTS	BEGIN_HINTS N0 COLOR_0 N1 COLOR_2 N2 COLOR_3 N10 COLOR_1 N15 COLOR_1 N17 COLOR_3 ... (37 tokens)
GRAPH	BEGIN_GRAPH N0 N16 N8 N38 N34 N17 N1 N13 N27 N34 N28 N10 ... (289 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 N34 COLOR_1 N44 ... (75 tokens)

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 13 rounds, 30 deductions,
60 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000018, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

18 hints of 48 nodes, then 13 rounds of simultaneous deduction. Hints: N0=COLOR_0 N1=COLOR_2 N2=COLOR_3 N10=COLOR_1 N15=COLOR_1 N17=COLOR_3 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND      N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 ...
  6 deduced
BEGIN_ROUND      N42 COLOR_3 N45 COLOR_1
  2 deduced
BEGIN_ROUND      N13 COLOR_0 N20 COLOR_0 N37 COLOR_2
  3 deduced
BEGIN_ROUND      N21 COLOR_2 N24 COLOR_3 N31 COLOR_2
  3 deduced
... 9 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N9, neighbours N25 N10 N29 N31 N18

their colours known before this round:

```
[0, 1, 2]  (3 = q-1 of the 4)

(3 known neighbours, 3 DISTINCT colours)

still unknown: N25 N31
```

only colour left: 3

```
so the trace emits  N9 COLOR_3
```

The model is never shown this reasoning -- only the node/colour pair.

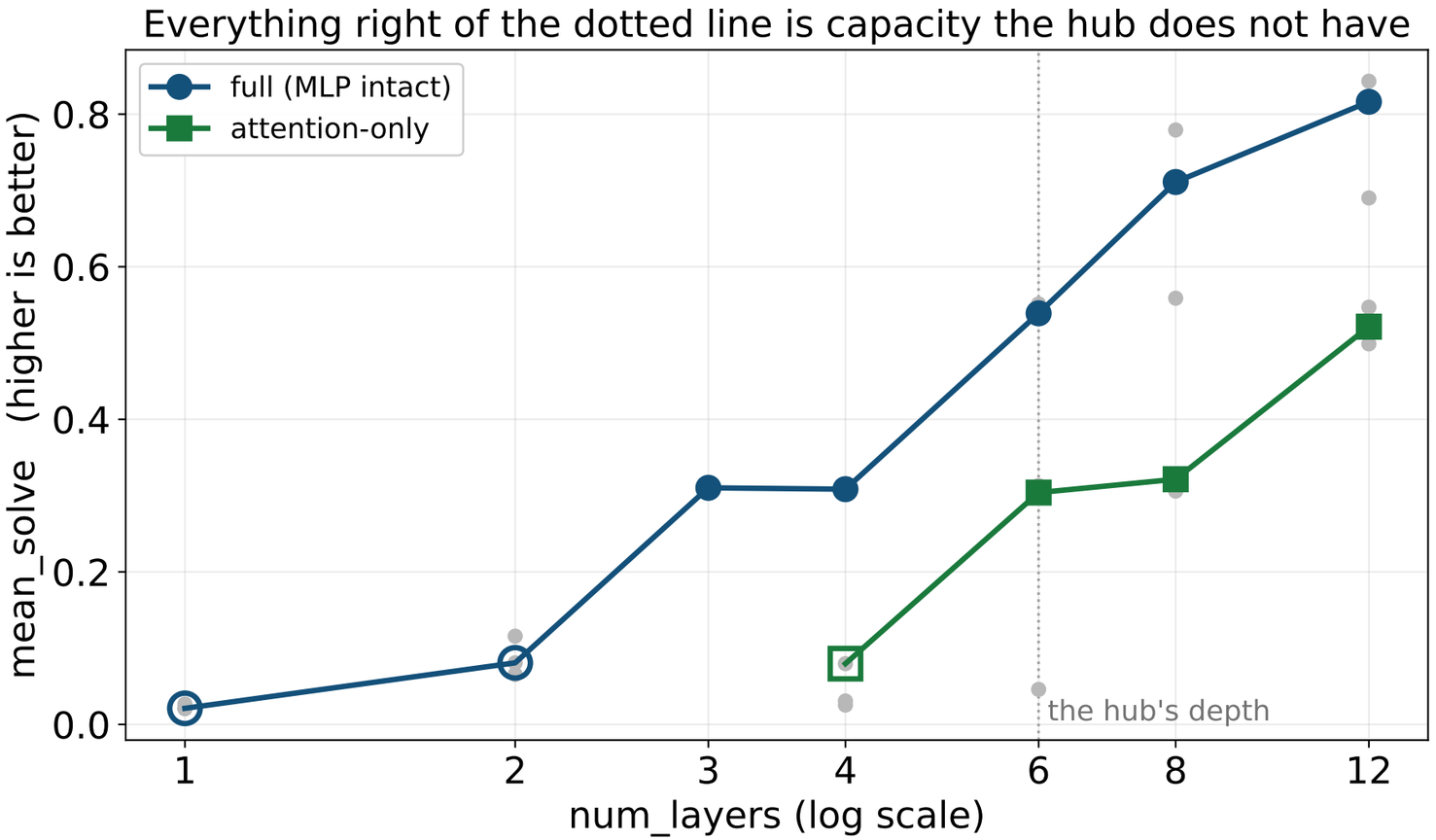
A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

N=48, degree 5, 4 colours, --shuffle-adj --fall: the task every cell in this deck is trained and scored on. The attention-only models have to produce this with no MLP anywhere in the stack, at L=8 and L=12 instead of the hub's L=6.

T=13 rounds, the 90th percentile of 40 draws from this config (range 1-16) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

Two depth ladders: the complete model, and the same model with no MLP

d_model 256, 4 heads, ff=1024 wherever an MLP exists; everything else at the hub's settings, 3 seeds per rung



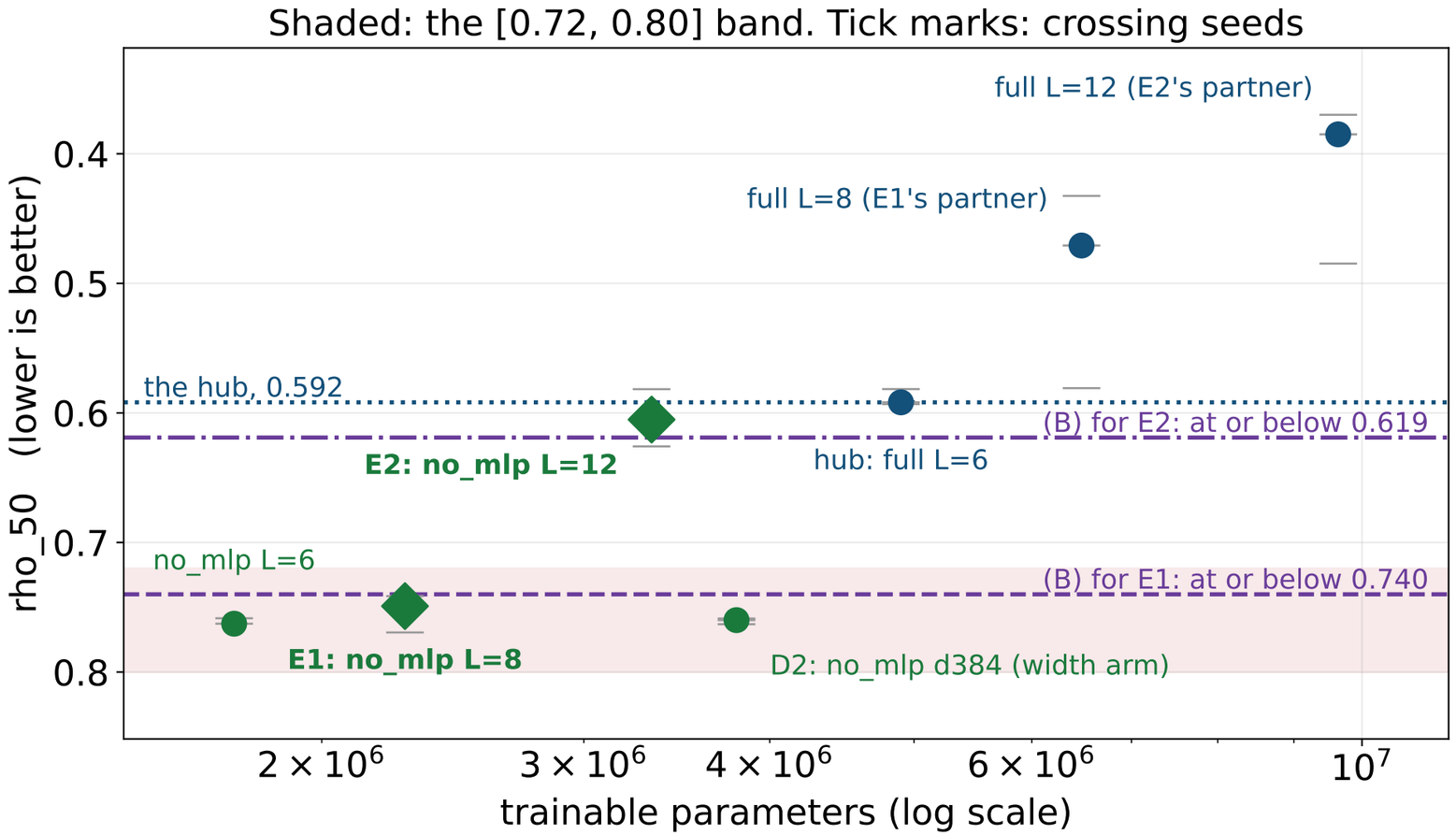
HOLLOW MARKER = never escaped the 1.64 plateau, so the cell is an optimization failure and its number is not a statement about depth.

	L	mean_solve	trainable	esc
full				
	1	0.0211	951,098	0/3
	2	0.0807	1,740,858	0/3
	3	0.3102	2,530,618	3/3
	4	0.3084	3,320,378	3/3
	6	0.5389	4,899,898	3/3
	8	0.7111	6,479,418	3/3
	12	0.8163	9,638,458	3/3
no_mlp				
	4	0.0796	1,218,106	0/3
	6	0.3039	1,746,490	2/3
	8	0.3214	2,274,874	3/3
	12	0.5214	3,331,642	3/3

Per-seed values in grey. The complete model's ladder is the reference the whole grid is built on: it descends with depth and keeps descending past L=6. The question stage E asks is whether the attention-only ladder does the same thing when it is given the same two extra depths, or whether it flattens wherever the pile-up at 0.76 sits. A hollow marker at the deep end answers neither question -- it says the cell did not train. Commit c04ecc9.

Where the two depth rungs landed, against predictions made first

Every line below was written into a committed sbatch header before the cell it describes finished




```
REGISTERED, THEN SCORED

E1   no_mlp L=8, 2,274,874
  (A)    rho_50 in [0.72, 0.8],
         mean_solve in [0.26, 0.38]
  (B)    rho_50 <= 0.74 -- joins the hub
  got    0.749 / 0.3214

E2   no_mlp L=12, 3,331,642
  (A)    rho_50 in [0.72, 0.8] again
  (B)    rho_50 <= 0.619 -- full L=8's rung
  got    0.605 / 0.5214

FALSIFIER OF (A), from the header:
EITHER rung with a median below 0.74
on >=2 of 3 crossing seeds ends the
barrier reading. No partial credit, no
'but the other rung held'. The overlap
with the tie-break clause is a conflict
IN THE REGISTRATION; page 6 reports it
rather than resolving it quietly.

Censored rungs cannot be placed on this
axis and are left out; all of them sit
BELOW the band, so leaving them out
understates the pile-up.
```

The two complete-model rungs at L=8 and L=12 are the paired controls: same 14-point grid, same 375k-step budget, same eval, same seeds. s* = 0.1484 is the seed-to-seed spread measured at fixed architecture, which is why (A) is a band and not a line, and why a gap smaller than one s* is not a result. Commit c04ecc9.

The two calls, and the roll-up across both axes

Transcribed from the sbatch headers into scripts/capacity_back.py, pinned by 15 tests before the cells landed

E1 -- HELD

(A) HOLDS ON THIS RUNG -- median rho_50 0.749 sits inside the registered [0.72, 0.80] window, on the 0.76 rung, with more of a resource than the hub and nothing to show for it.

E2 -- FALSIFIED

(A) FALSIFIED BY THIS RUNG ALONE -- median rho_50 0.605 is below the tie-break floor 0.72 on 3 of 3 crossing seeds. Capacity given back DOES cross the band, so 0.76 is not a barrier and the pile-up is reading (B).

ACROSS D2, E1 AND E2 -- the scorer's own roll-up, not the deck's

VERDICT: (A) BARRIER IS FALSIFIED, by E2. An ablated rung handed more capacity than the hub crossed below the tie-break floor. The pile-up at 0.76 is reading (B) CAPACITY: those ablations cost about the same at that budget, and there is no floor. One rung is enough -- the registration says there is no partial credit and no 'but the other arm held'. Do not write 'the MLP is load-bearing in kind' anywhere.

Trainability is checked first and outranks everything: 0 plateau escapes means 'did not train in 375k steps' and is NEVER evidence in either direction -- the trap stage C's P2 fell into. Then fewer than three seeds, then fewer than two crossing seeds, then a median below 0.72, then the tie-break region up to 0.740, then the band's upper edge at 0.80. A rung that never got the chance to break the barrier reading is not counted as one that held it. Commit c04ecc9.

What this does not settle, and what is still running

Stage G (51194118) doubles the budget; stage H (51223007) tries to give the size back

THE BUDGET ARM, REGISTERED BEFORE ITS FIRST CELL FINISHED

Every rung in this deck ran 50 epochs. Stage G reruns attention-only at d128 and d256 for 100, three seeds each, with an epoch-50 reproduction gate: the first 50 epochs must match the frozen 50-epoch twins to within 0.01 nats, or the cells are not twins and nothing is comparable. All three in-flight cells passed that gate at epoch 16-23 with a worst deviation of 0.008. It is the only way a rung written off here as 'did not train' can come back.

AND THE PREDICTION FOR IT IS THAT IT FAILS

scripts/escape_timing.py Q3 measured where the 14 late plateau escapes actually came from: all 14 crossed from a train NLL of 1.640-1.652, and 13 of the 14 were within five epochs of the plateau at their own prior rate. There is no knee. Extrapolating the 39 never-escaping cells at their own last-20% slope puts the median 87 epochs short, and only 10 of 39 within another 50 -- none of them a stage G cell. So doubling the budget is registered to NOT rescue the d128 rung, which contradicts stage G's own header, and that contradiction is on record.

THE STANDING MINIMUM MOVED, AND IT MOVED THE WRONG WAY ON SIZE

Best on this eval at any size, same grid and budget: the complete model at L=12, rho_50 0.385 at 9,638,458 trainable -- nothing here approaches it and nothing was trying to. The brief is the SMALLEST architecture that still learns direct deduction, and the replacement rule was fixed in stage H's header before E2 was scored: 3 of 3 crossing seeds AND a median at or below 0.74. E2 is the first rung to satisfy it, so the standing minimum is now attention-only L=12, 3,331,642 trainable at rho_50 0.605, taking it from attention-only L=6, 1,746,490 at 0.763 on 2 of 3 seeds: rho_50 fell 0.158 (1.06 s*) and the size ROSE 1.91x.

The resolution limit stands whatever the calls on page 6 say: rung_collapse found no DETECTABLE separation between architecture labels inside the band (p = 0.569, positive control p = 0.045 on a separation 26x larger), which is not the same claim as 'the architectures are equivalent'. Commit c04ecc9.